

Linear transformations and matrices

CHAPTER 3

Linear transformations and matrices

Where matrices finally make sense: they move space.

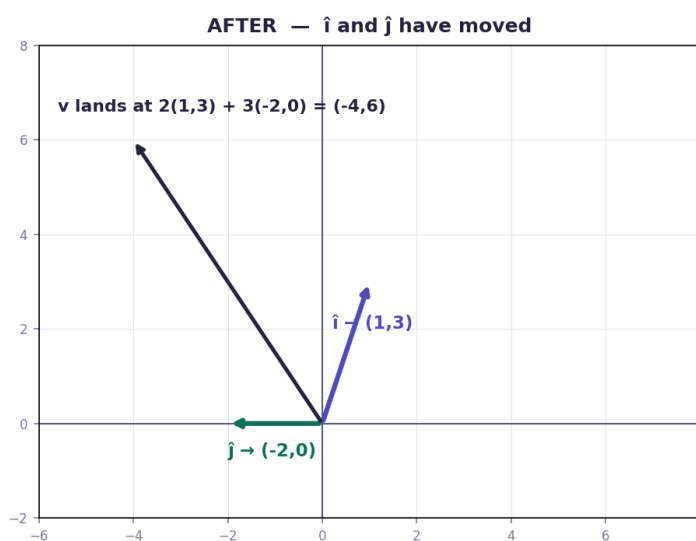
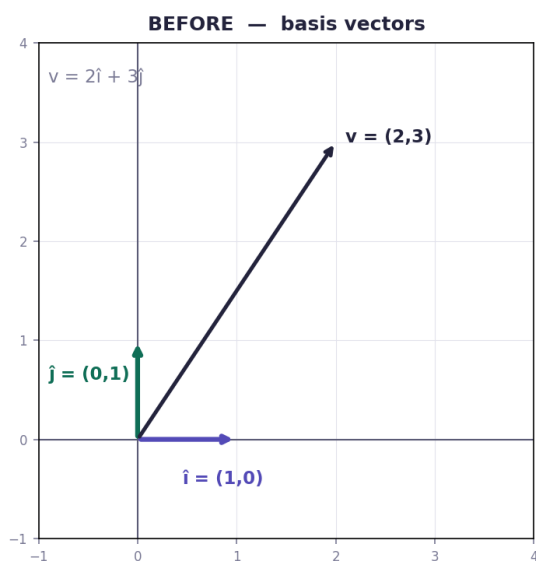
THE IDEA

A **linear transformation** moves every point in space, but keeps grid lines straight, parallel and evenly spaced, with the origin fixed. The remarkable fact: to know where *every* vector goes, you only need to know where **$\hat{i}$ and $\hat{j}$** go.

The whole chapter in one line: the columns of a matrix are where $\hat{i}$ and $\hat{j}$ land.

READING A MATRIX

Matrix part	Meaning
Column 1	Where $\hat{i} = (1, 0)$ lands
Column 2	Where $\hat{j} = (0, 1)$ lands
Applying it	input $(x, y) \rightarrow x \cdot (\text{column 1}) + y \cdot (\text{column 2})$, then add



WHY IT WORKS

The input $(2, 3)$ literally means "2 steps along $\hat{i}$, 3 steps along $\hat{j}$ ". After the transformation, $\hat{i}$ and $\hat{j}$ have moved — so the answer is 2 of *where $\hat{i}$ went* plus 3 of *where $\hat{j}$ went*. Nothing more.

WORKED EXAMPLE

CHAPTER 3 — WORKED EXAMPLES

matrix columns = where $\hat{i}$ and $\hat{j}$ land

RULE

$$\text{input } (x, y) \rightarrow x \cdot (\text{column 1}) + y \cdot (\text{column 2})$$

EXAMPLE

$$\hat{i} \rightarrow (1, 3) \quad \hat{j} \rightarrow (-2, 0) \quad \text{input } (-2, 4)$$

$$\begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix} \quad x = -2, \quad y = 4$$

$$-2 \times (1, 3) = (-2, -6)$$

$$4 \times (-2, 0) = (-8, 0)$$

$$\text{add : } (-2 + (-8), -6 + 0) = (-10, -6)$$

This is matrix $\times$ vector — the core operation of every neural network.

TEST — Chapter 3

1. $\hat{i} \rightarrow (2, 0), \hat{j} \rightarrow (0, 3)$. Where does $(1, 2)$ land? $\rightarrow (2, 6)$
2. $\hat{i} \rightarrow (1, 3), \hat{j} \rightarrow (-2, 0)$. Where does $(-2, 4)$ land? $\rightarrow (-10, -6)$
3. $\hat{i} \rightarrow (1, -2), \hat{j} \rightarrow (3, 1)$. Where does $(2, -3)$ land? $\rightarrow (-7, -7)$

FORMULAS FROM THIS CHAPTER

Matrix columns

column 1 = where $\hat{i}$ lands, column 2 = where $\hat{j}$ lands

Apply to vector

$x \cdot (\text{col 1}) + y \cdot (\text{col 2})$